

Multiple Choice (10 marks)

1. Polarization is a property of
 - (a) mechanical waves
 - (b) sound waves
 - (c) all waves
 - (d) None of these
 - (e) radio waves
2. Which of these does NOT belong in the family of electromagnetic waves?
 - (a) Sound
 - (b) Ultraviolet waves
 - (c) Infrared waves
 - (d) Microwaves
 - (e) X-rays
3. The magnitude of the gravitational force on a satellite is constant if the orbit is
 - (a) oval
 - (b) spiral
 - (c) elliptical
 - (d) All of these
 - (e) rectangular
4. Two thin convex lenses of focal lengths f_1 and f_2 are separated by a distance equal to $2 f_2$. If $f_1 = 3 f_2$, what is the focal length of the combination?
 $(3 f_2) / 2$
 - (a) $(f_1 + f_2) / 2$
 - (b) $(2 f_2) / 3$
 - (c) $2 f_2$
 - (d) $(3 f_2) / 3$
5. According to the BCS theory, the attraction between Cooper pairs in a superconductor is due to
 - (a) Magnetic attraction
 - (b) vacuum polarization

- (c) interactions with the ionic lattice
- (d) Gravitational pull
- (e) Electrostatic attraction

True / False (10 marks)

Answer True or False

6. True or False: Relativistic equations for time dilation, length contraction, and relativistic momentum hold true at all speeds.
7. True or False: Applying 120 volts to a 240-ohm resistor results in a current of 0.4 amperes.
8. True or False: The total spin quantum number of the electrons in the ground state of neutral nitrogen ($Z = 7$) is 1.
9. True or False: Jupiter's internal heat is mainly produced by nuclear fusion processes occurring in its core.
10. True or False: The buoyant force acting on an object submerged in a fluid is not affected by the depth of the fluid.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the thermal expansion considerations for a modern steel bridge spanning two miles, subjected to temperature variations from -40°C to $+110^{\circ}\text{C}$, and calculate the expected expansion.
11. Calculate the total electric charge in coulombs of 75.0 kg of electrons, and discuss the significance of your result in the context of charge conservation and its implications in physics.

End of Paper